

Final Project ISA491/591

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Load libraries needed to run code

Data preparation

```
df <- read.csv("Congestion-1.csv", stringsAsFactors = TRUE)
```

```
head(df)
```

```
##           Time Date Day.of.the.week CarCount BikeCount BusCount TruckCount
Total
## 1 12:00:00 AM   10           Tuesday      31         0         4         4
39
## 2 12:15:00 AM   10           Tuesday      49         0         3         3
55
## 3 12:30:00 AM   10           Tuesday      46         0         3         6
55
## 4 12:45:00 AM   10           Tuesday      51         0         2         5
58
## 5  1:00:00 AM   10           Tuesday      57         6        15        16
94
## 6  1:15:00 AM   10           Tuesday      44         0         5         4
53

##      Congestion
## 1             No
## 2             No
## 3             No
## 4             No
## 5             No
## 6             No
```

```
summary(df)
```

```
##           Time           Date      Day.of.the.week      CarCount
## 1:00:00 AM: 31   Min.      : 1.00   Friday       :383   Min.      : 6.00
## 1:00:00 PM: 31   1st Qu.: 8.00   Monday       :384   1st Qu.: 19.00
## 1:15:00 PM: 31   Median :16.00   Saturday    :384   Median : 64.00
## 1:30:00 AM: 31   Mean      :15.99   Sunday      :383   Mean      : 68.72
## 1:30:00 PM: 31   3rd Qu.:24.00   Thursday    :476   3rd Qu.:107.00
## 1:45:00 AM: 31   Max.      :31.00   Tuesday     :478   Max.      :180.00
## (Other)      :2780                      Wednesday:478

##      BikeCount      BusCount      TruckCount      Total      Congestion
## Min.      : 0.00   Min.      : 0.00   Min.      : 0.00   Min.      : 21.0   No :1967
## 1st Qu.: 5.00   1st Qu.: 1.00   1st Qu.: 6.00   1st Qu.: 55.0   Yes: 999
## Median :12.00   Median :12.00   Median :14.00   Median :109.0
## Mean      :14.92   Mean      :15.27   Mean      :15.32   Mean      :114.2
## 3rd Qu.:22.00   3rd Qu.:25.00   3rd Qu.:23.00   3rd Qu.:164.0
## Max.      :70.00   Max.      :50.00   Max.      :40.00   Max.      :279.0
##
```

Introduction

Completing this data analysis had the objective of developing an accurate model to predict traffic congestion based on vehicle counts. There were 2,966 observations that were used in the dataset and they were recorded every 15 minutes. Within the analysis two predictive models were built and then compared, a logistic regression and random forest. The results show that a random forest model had the more accurate predictive performance, making it the better choice for real time traffic monitoring.

Data cleaning

The dataset was clean with zero missing values found using `sum(is.na(df))`. The average total vehicle count was 114.2 with a maximum of 279. Out of cars, bikes, buses, and trucks; cars were the primary vehicle count in the traffic mix with a mean count of 68.7 which was a much higher mean count than any other vehicle. Out of the total 2,966 observations, 999 were labeled as congested (Yes) and 1,967 as not congested (No).

There were four variables that were engineered to give the models better context. The variable Hour was extracted from the timestamp. The binary IsPeak variable was reassigned to morning and evening rush hours. IsWeekend flagged Saturday and Sunday observations. Vehicle and ratio variables were also found in order to capture the composition of traffic.

```
# Check for missing values
```

```
sum(is.na(df))
```

```
## [1] 0
```

```
df <- na.omit(df)
```

```
# Extract hour from Time and create binary flags
```

```
df$Hour <- as.integer(format(strptime(df$Time, "%I:%M:%S %p"), "%H"))
```

```
# Ordered day factor
```

```
df$Day.of.the.week <- factor(df$Day.of.the.week,
```

```
  levels = c("Monday", "Tuesday", "Wednesday",
```

```
             "Thursday", "Friday", "Saturday", "Sunday"))
```

```
# Response variable
```

```
df$Congestion <- factor(df$Congestion, levels = c("No", "Yes"))
```

```
# Define specific hour sets
```

```
peak_hours <- c(7, 8, 9, 16, 17, 18, 19)
```

```
weekend_days <- c("Saturday", "Sunday")
```

```
# Assign binary flags
```

```
df$IsPeak <- as.integer(df$Hour %in% peak_hours)
```

```
df$IsWeekend <- as.integer(df$Day.of.the.week %in% weekend_days)
```

```
# Calculate ratios of vehicle types
```

```
df$CarRatio <- df$CarCount / (df$Total + 1)
```

```
df$BikeRatio <- df$BikeCount / (df$Total + 1)
```

```
df$BusRatio <- df$BusCount / (df$Total + 1)
```

```
df$TruckRatio <- df$TruckCount / (df$Total + 1)
```

```
# Drop unused columns
```

```
df$Time <- NULL
```

```
df$Date <- NULL
```

```
summary(df)
```

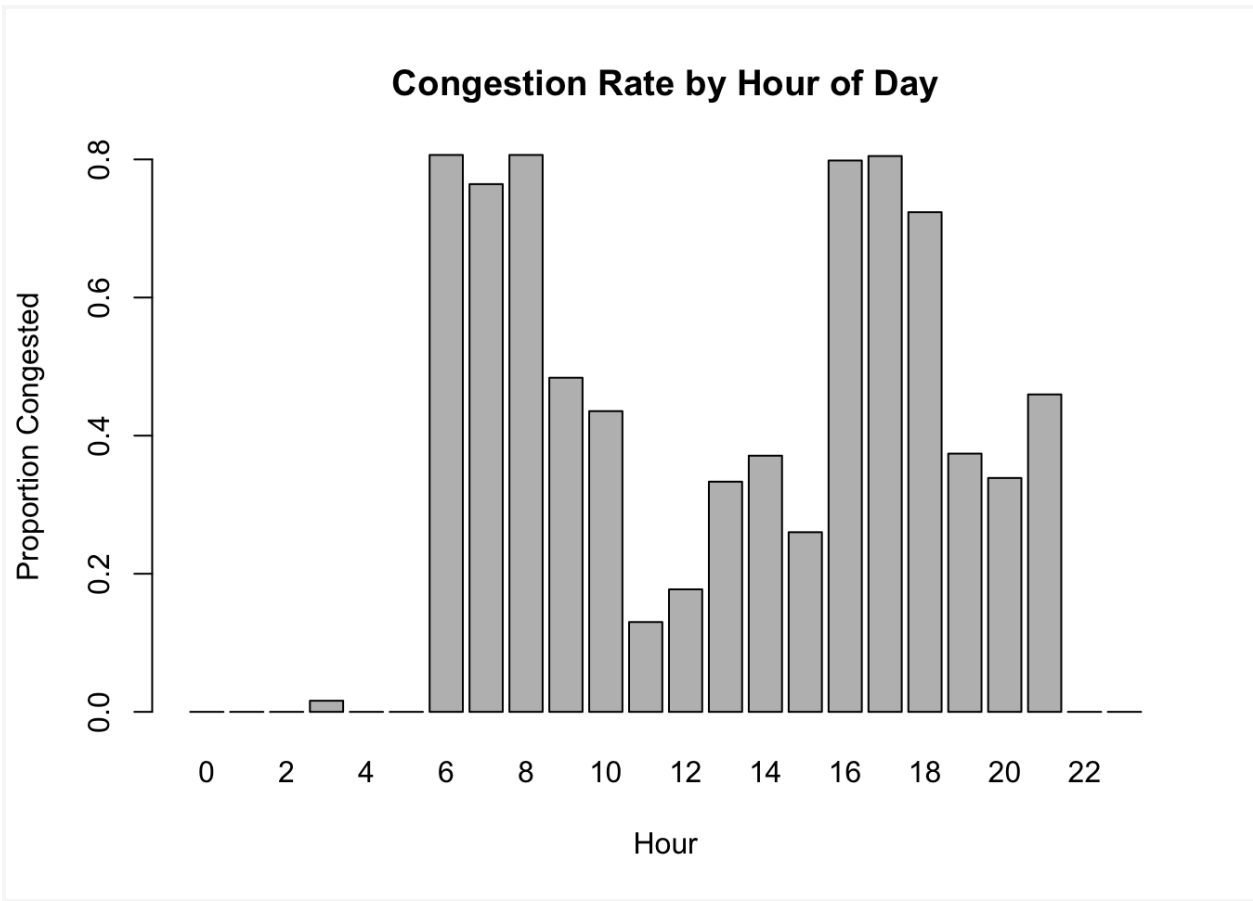
```
##   Day.of.the.week   CarCount      BikeCount      BusCount
## Monday      :384   Min.      : 6.00   Min.      : 0.00   Min.      : 0.00
## Tuesday     :478   1st Qu.: 19.00   1st Qu.: 5.00   1st Qu.: 1.00
## Wednesday:478   Median : 64.00   Median :12.00   Median :12.00
## Thursday :476   Mean      : 68.72   Mean      :14.92   Mean      :15.27
## Friday      :383   3rd Qu.:107.00   3rd Qu.:22.00   3rd Qu.:25.00
## Saturday   :384   Max.      :180.00   Max.      :70.00   Max.      :50.00
## Sunday      :383
##   TruckCount      Total      Congestion      Hour      IsPeak
## Min.      : 0.00   Min.      : 21.0   No :1967   Min.      : 0.00   Min.      :0.0000
## 1st Qu.: 6.00   1st Qu.: 55.0   Yes: 999   1st Qu.: 5.25   1st Qu.:0.0000
## Median :14.00   Median :109.0
## Mean      :15.32   Mean      :114.2
## 3rd Qu.:23.00   3rd Qu.:164.0
## Max.      :40.00   Max.      :279.0
## Max.      :23.00   Max.      :1.0000
##
##   IsWeekend      CarRatio      BikeRatio      BusRatio
## Min.      :0.0000   Min.      :0.07692   Min.      :0.00000   Min.      :0.00000
## 1st Qu.:0.0000   1st Qu.:0.41667   1st Qu.:0.06667   1st Qu.:0.02564
## Median :0.0000   Median :0.57961   Median :0.10748   Median :0.10018
## Mean      :0.2586   Mean      :0.54200   Mean      :0.11579   Mean      :0.10843
## 3rd Qu.:1.0000   3rd Qu.:0.66488   3rd Qu.:0.16069   3rd Qu.:0.16759
## Max.      :1.0000   Max.      :0.96396   Max.      :0.48387   Max.      :0.62712
```

```
##  
##      TruckRatio  
##  Min.      :0.00000  
## 1st Qu.:0.04104  
##  Median :0.13889  
##   Mean   :0.22145  
## 3rd Qu.:0.36655  
##   Max.   :0.78000  
##
```

Using visualizations of bar plots of congestion rate by hour and by day of week, structure of the data was revealed. Congestion was heavily concentrated during peak commuting hours. This confirms that Hour and IsPeak are meaningful predictors. Friday was an unusual day and the logistic regression assigned it as a statistically significant positive coefficient (Estimate = 3.478, $p < 0.001$). This suggests that Friday traffic behaves differently from the rest of the week.

Congestion by hour

```
hour_tab <- table(df$Hour, df$Congestion)  
hour_pct <- prop.table(hour_tab, margin = 1)  
barplot(hour_pct[, "Yes"],  
        main = "Congestion Rate by Hour of Day",  
        xlab = "Hour",  
        ylab = "Proportion Congested",  
        names.arg = rownames(hour_pct))
```



```
# Congestion by day of the week
```

```
day_tab <- table(df$Day.of.the.week, df$Congestion)
```

```
day_pct <- prop.table(day_tab, margin = 1)
```

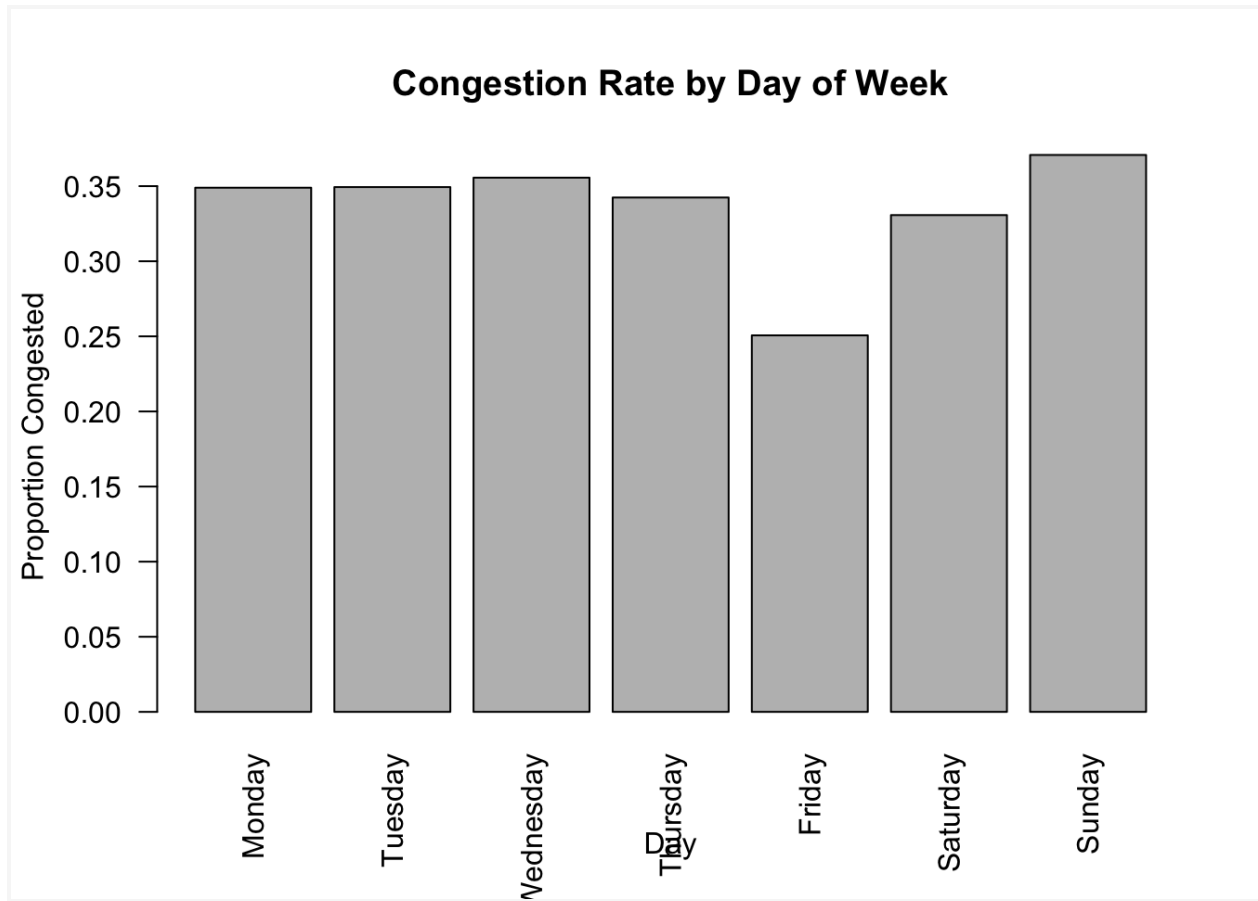
```
barplot(day_pct[, "Yes"],
```

```
        main = "Congestion Rate by Day of Week",
```

```
        xlab = "Day",
```

```
        ylab = "Proportion Congested",
```

```
        las = 2)
```



Vehicle counts and congestion

```
par(mfrow = c(2, 3))
```

```
boxplot(CarCount ~ Congestion, data = df, main = "Car")
```

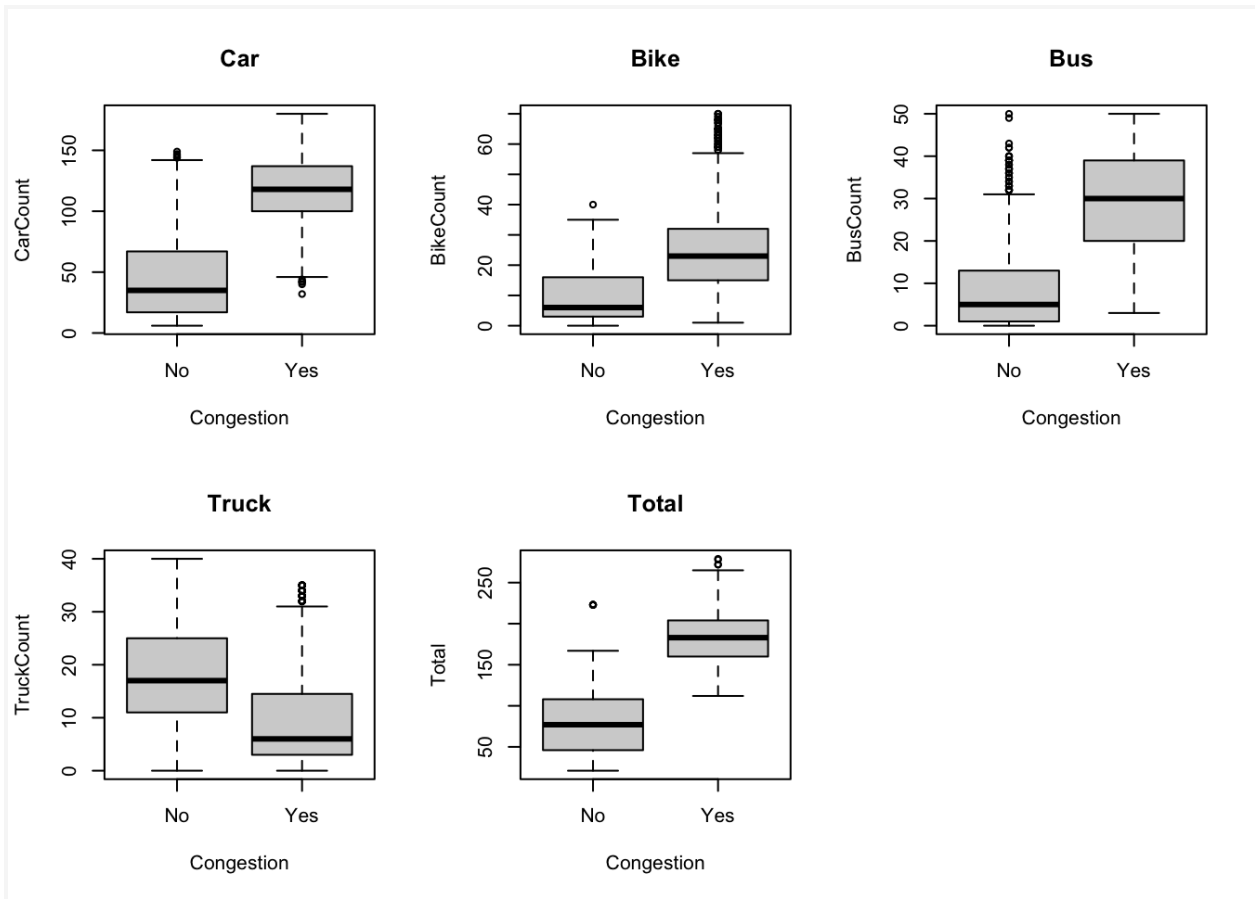
```
boxplot(BikeCount ~ Congestion, data = df, main = "Bike")
```

```
boxplot(BusCount ~ Congestion, data = df, main = "Bus")
```

```
boxplot(TruckCount ~ Congestion, data = df, main = "Truck")
```

```
boxplot(Total ~ Congestion, data = df, main = "Total")
```

```
par(mfrow = c(1, 1))
```



The data was split 80/20 into a training set with 2,373 rows and a test set of 593 rows. Both models were trained on the same split using `set.seed(123)`. The logistic regression identified `CarCount` ($p < 0.001$), `BikeCount` ($p < 0.01$), and `TruckCount` ($p < 0.001$) as significant predictors, while `BusCount` was not significant ($p = 0.81$). This significance suggests that the buses that are running on fixed schedules do not add any meaningful predictive information. `CarRatio`, `BikeRatio`, `BusRatio`, and `TruckRatio` all being highly significant proves that the mix of vehicle types on the road matters just as much as the total number of vehicles. The variable `IsWeekend` was dropped because it had perfect collinearity with `Day.of.the.week`. The random forest model was trained with 300 trees and `mtry = 3`. The error rate was 2.53% with only 60 misclassifications out of 2,373 training observations.

Model Development

```
set.seed(123)

trn.indx <- sample(1:nrow(df), size = round(0.8 * nrow(df)), replace = FALSE)

df.trn <- df[trn.indx, ]
df.tst  <- df[-trn.indx, ]

cat("Training set:", nrow(df.trn), "rows\n")
```



```
## Training set: 2373 rows
```

```
cat("Test set      :", nrow(df.tst), "rows\n")
```

```
## Test set       : 593 rows
```

The performance of the logistic regression and random forest: Metric Logistic Regression Random Forest Accuracy 96.8% 98.8% Misclassification Rate 3.2% 1.2% AUC 0.9926 0.9995

The logistic regression confusion matrix showed 14 false positives and 5 false negatives on the test set. The random forest reduced this to 4 false positives and 3 false negatives. The AUC scores also favor the random forest model, having an AUC of 0.9995 indicates a near perfect discrimination between congested and non congested conditions, whereas the logistic regression had an AUC of 0.9926.

Logistic Regression

```
fit.log <- glm(Congestion ~ CarCount + BikeCount + BusCount + TruckCount +  
              Hour + Day.of.the.week + IsWeekend + IsPeak +  
              CarRatio + BikeRatio + BusRatio + TruckRatio,  
              data = df.trn, family = "binomial")
```

```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```
summary(fit.log)
```

```
##
```

```
## Call:
```

```
## glm(formula = Congestion ~ CarCount + BikeCount + BusCount +  
##      TruckCount + Hour + Day.of.the.week + IsWeekend + IsPeak +  
##      CarRatio + BikeRatio + BusRatio + TruckRatio, family = "binomial",  
##      data = df.trn)
```

```
##
```

```
## Coefficients: (1 not defined because of singularities)
```

```
##              Estimate Std. Error z value Pr(>|z|)  
## (Intercept)   -6.764e+03  9.859e+02  -6.861 6.86e-12 ***  
## CarCount      -1.360e-01  3.226e-02  -4.215 2.50e-05 ***  
## BikeCount     -1.908e-01  7.277e-02  -2.621 0.008755 **  
## BusCount      -1.931e-02  7.850e-02  -0.246 0.805661
```

```
## TruckCount          -7.328e-01  1.480e-01  -4.951  7.39e-07  ***
## Hour                3.369e-02  2.452e-02   1.374  0.169439
## Day.of.the.weekTuesday -5.390e-02  3.739e-01  -0.144  0.885377
## Day.of.the.weekWednesday -4.314e-01  3.743e-01  -1.153  0.249100
## Day.of.the.weekThursday -3.460e-01  4.029e-01  -0.859  0.390433
## Day.of.the.weekFriday   3.478e+00  9.946e-01   3.497  0.000471  ***
## Day.of.the.weekSaturday -6.224e-01  3.846e-01  -1.618  0.105606
## Day.of.the.weekSunday   9.643e-02  4.040e-01   0.239  0.811337
## IsWeekend              NA          NA          NA          NA
## IsPeak                -1.623e-01  2.446e-01  -0.663  0.507038
## CarRatio              6.823e+03  9.962e+02   6.849  7.44e-12  ***
## BikeRatio            6.835e+03  9.977e+02   6.851  7.35e-12  ***
## BusRatio              6.834e+03  1.001e+03   6.827  8.69e-12  ***
## TruckRatio            6.942e+03  1.014e+03   6.846  7.57e-12  ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 3053.08  on 2372  degrees of freedom
## Residual deviance:  587.92  on 2356  degrees of freedom
## AIC: 621.92
##
## Number of Fisher Scoring iterations: 13
```

Random Forest

```
set.seed(123)
```

```
fit.rf <- randomForest(Congestion ~ Hour + Day.of.the.week + IsWeekend + IsPeak
+
  CarCount + BikeCount + BusCount + TruckCount,
```

```

data = df.trn,

ntree = 300,

mtry = 3

)

print(fit.rf)

##

## Call:

##  randomForest(formula = Congestion ~ Hour + Day.of.the.week +      IsWeekend
+ IsPeak + CarCount + BikeCount + BusCount + TruckCount,      data = df.trn,
ntree = 300, mtry = 3)

##              Type of random forest: classification

##              Number of trees: 300

## No. of variables tried at each split: 3

##

##          OOB estimate of  error rate: 2.53%

## Confusion matrix:

##          No Yes class.error
## No  1532  26  0.01668806
## Yes   34 781  0.04171779

```

Model Validation and Performance

```
# Logistic regression predictions
```

```

prob.log <- predict(fit.log, newdata = df.tst, type = "response")

pred.log <- factor(ifelse(prob.log > 0.5, "Yes", "No"), levels = c("No",
"Yes"))

```

```
# Random forest predictions
```

```

prob.rf <- predict(fit.rf, newdata = df.tst, type = "prob")

pred.rf <- predict(fit.rf, newdata = df.tst, type = "class")

```

```
# Performance metrics
```

```

cm_log <- confusionMatrix(pred.log, df.tst$Congestion)
cm_rf  <- confusionMatrix(pred.rf, df.tst$Congestion)

# Errors
err.log <- 1 - mean(df.tst$Congestion == pred.log)
err.rf  <- 1 - mean(df.tst$Congestion == pred.rf)

cat("Logistic Accuracy:", round(cm_log$overall['Accuracy'], 4), "\n")
## Logistic Accuracy: 0.968
cat("Random Forest Accuracy:", round(cm_rf$overall['Accuracy'], 4), "\n")
## Random Forest Accuracy: 0.9882
cat("Logistic misclassification rate:", round(err.log, 4), "\n")
## Logistic misclassification rate: 0.032
cat("Random Forest misclassification:", round(err.rf, 4), "\n")
## Random Forest misclassification: 0.0118

```

Confusion Matrices

```

table(Predicted = pred.log, Actual = df.tst$Congestion)

##           Actual
## Predicted  No Yes
##      No   395   5
##      Yes   14 179

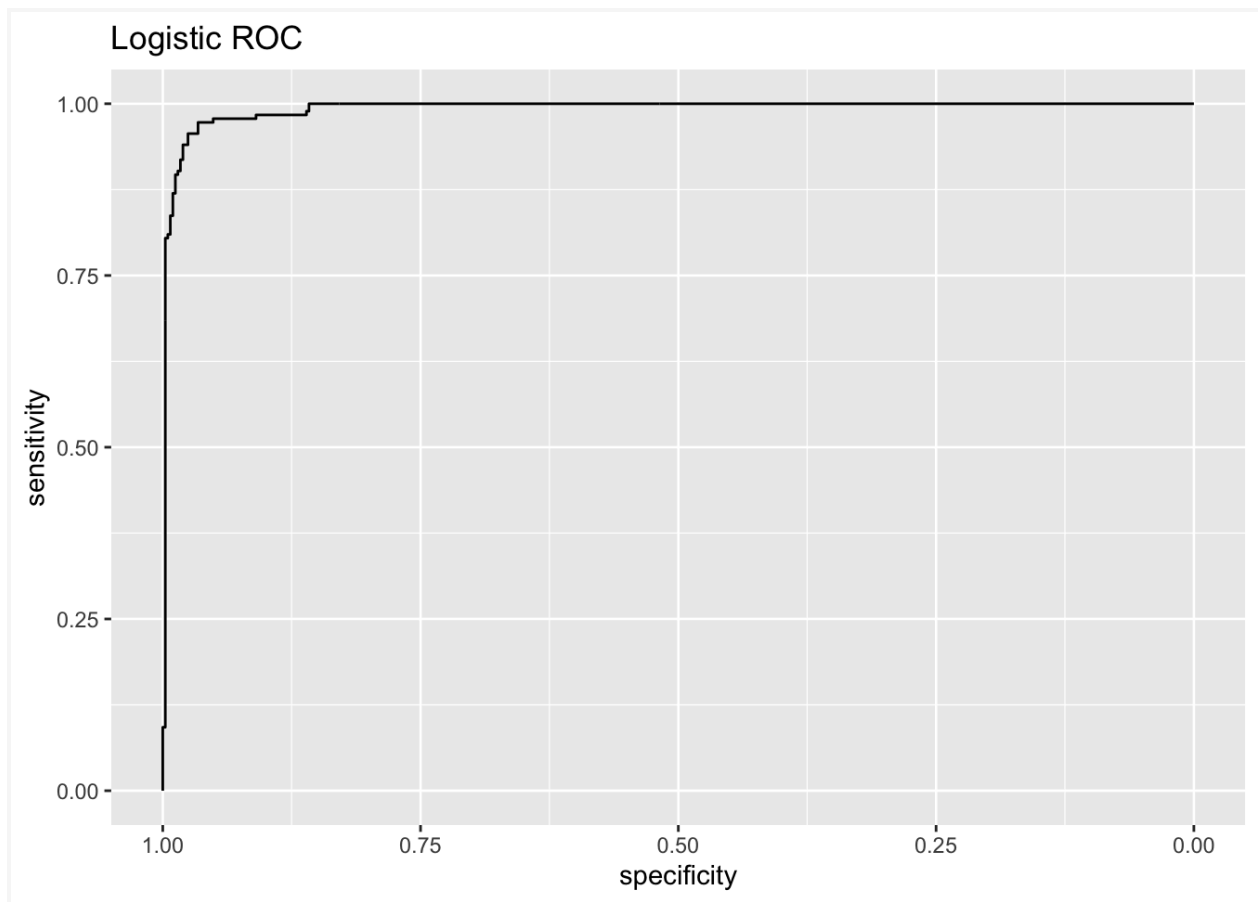
table(Predicted = pred.rf, Actual = df.tst$Congestion)

##           Actual
## Predicted  No Yes
##      No   405   3
##      Yes    4 181

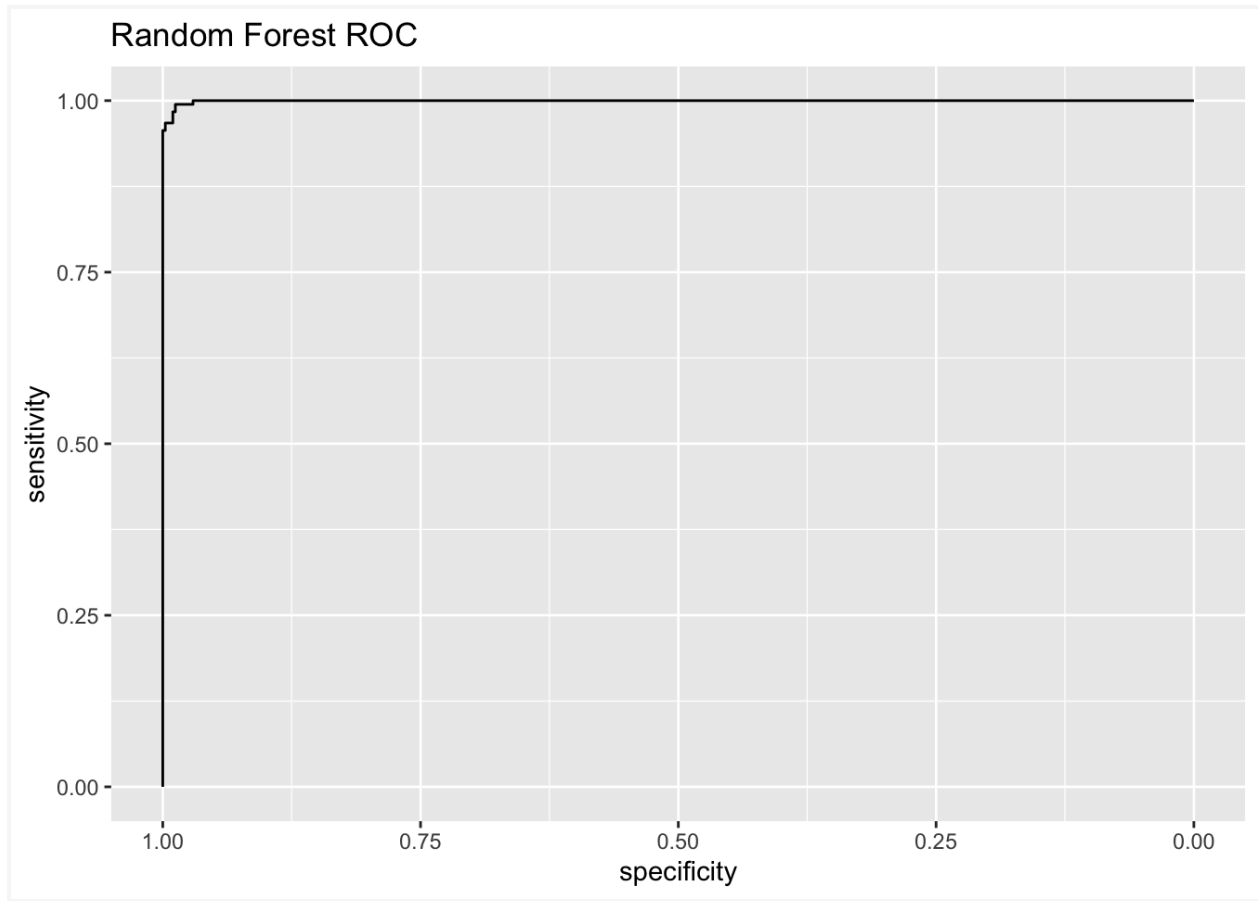
```

ROC Curves and AUC

```
r.log <- roc(df.tst$Congestion, prob.log, levels = c("No", "Yes"))  
  
## Setting direction: controls < cases  
  
r.rf <- roc(df.tst$Congestion, prob.rf[, "Yes"], levels = c("No", "Yes"))  
  
## Setting direction: controls < cases  
  
ggroc(r.log) + ggtitle("Logistic ROC")
```



```
ggroc(r.rf) + ggtitle("Random Forest ROC")
```



```
cat("Logistic AUC:", round(auc(r.log), 4), "\n")
```

```
## Logistic AUC: 0.9926
```

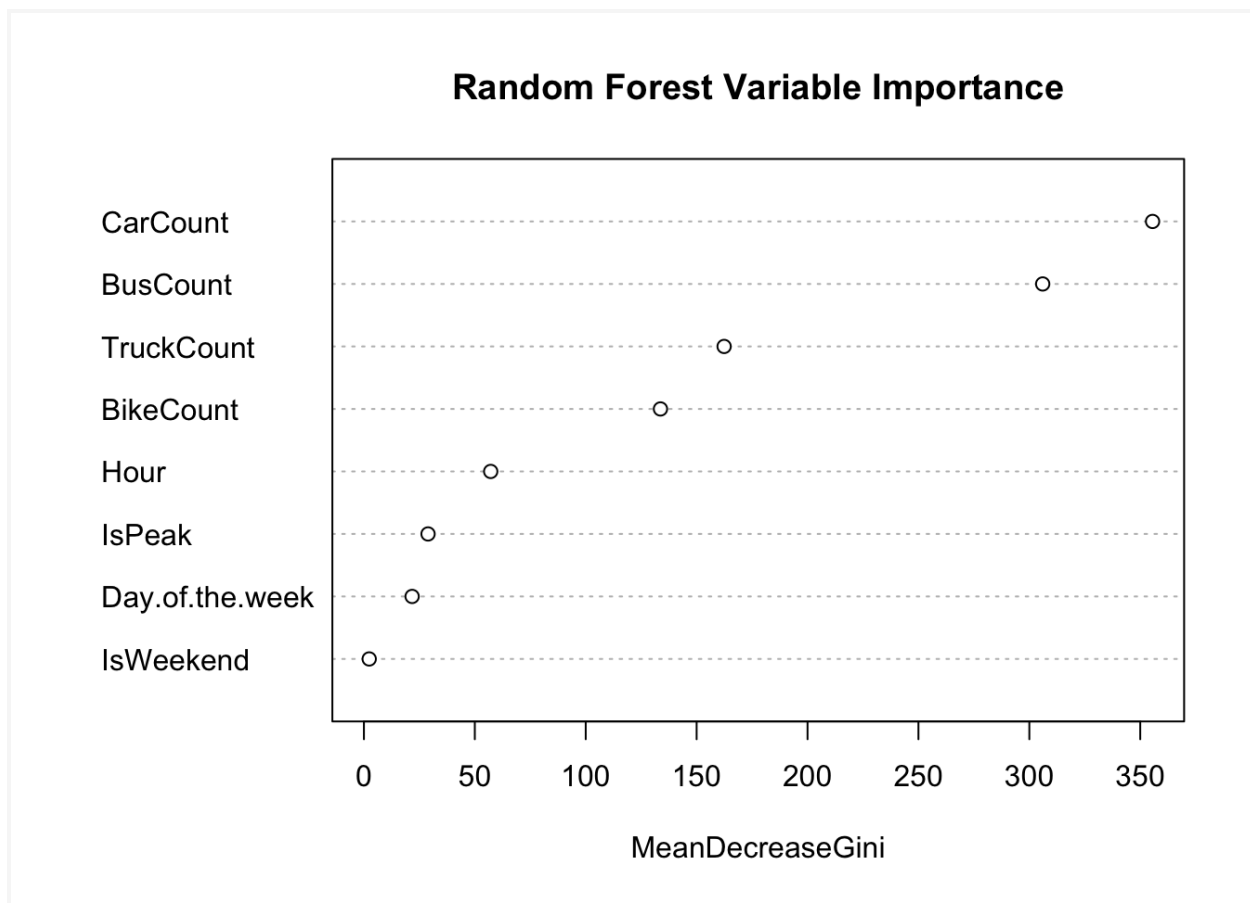
```
cat("Random Forest AUC:", round(auc(r.rf), 4), "\n")
```

```
## Random Forest AUC: 0.9995
```

The varImpPlot from the random forest model showed that Total vehicle volume was the strongest predictor of congestion followed by CarCount and Hour.TruckCount was also highly ranked probably because trucks occupy more road space that in turn slows traffic down.

Variable Importance

```
varImpPlot(fit.rf, main = "Random Forest Variable Importance")
```



Code for quiz

Predictions

```
df.score <- read.csv("Congestion_score-1.csv", stringsAsFactors = TRUE)

df.score$Hour <- as.integer(format(strptime(df.score$Time, "%I:%M:%S %p"),
"%H"))

df.score$Day.of.the.week <- factor(df.score$Day.of.the.week,
  levels =
c("Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"))

peak_hours <- c(7, 8, 9, 16, 17, 18, 19)

weekend_days <- c("Saturday", "Sunday")

df.score$IsPeak <- as.integer(df.score$Hour %in% peak_hours)

df.score$IsWeekend <- as.integer(df.score$Day.of.the.week %in% weekend_days)
```

```

df.score$CarRatio <- df.score$CarCount / (df.score$Total + 1)
df.score$BikeRatio <- df.score$BikeCount / (df.score$Total + 1)
df.score$BusRatio <- df.score$BusCount / (df.score$Total + 1)
df.score$TruckRatio <- df.score$TruckCount / (df.score$Total + 1)
pred.score <- predict(fit.rf, newdata = df.score, type = "class")
print(pred.score)

##      1      2      3      4      5      6      7      8      9     10
##   No   No Yes  Yes   No   No Yes   No   No Yes
## Levels: No Yes

```

Final Report

In conclusion, this analysis was successful in producing a robust predictive model and the random forest model is the clear recommendation for predicting traffic congestion. With 98.8% accuracy, an AUC of 0.9995, and only 7 misclassifications, it outperforms logistic regression on each metric. The most critical drivers of congestion are total vehicle volume, car count, and time of day.